

Sam Kay AMIMA

MATHEMATICIAN · MUSICIAN · PRIVATE TUTOR

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Personal Profile

As a highly motivated and dedicated student with a deep passion for learning and personal growth, I consistently go above and beyond in my studies. I am recognised for my exceptional work ethic and efficient time management skills, which have enabled me to consistently meet deadlines and attain outstanding academic results. I possess an unwavering aspiration to succeed and am always endeavouring to elevate myself to new heights of accomplishment.

I am agreeable, enthusiastic and a clear communicator. A skill I am most proud of is the ability to convey complex ideas with a flexible vocabulary, tailoring conversations to a wide variety of audiences.

Education

Durham University

Doctor of Philosophy (PhD) in Mathematical Sciences

Durham, UK

Oct 2025 - Current

- Working towards a PhD under the supervision of Dr Peter Wyper.
- Researching structures in the solar wind, which falls under the realm of magnetohydrodynamics (MHD).

Durham University

Master of Mathematics (MMath)

Durham, UK

Sept 2021 - July 2025

- Graduated with Master's degree in mathematics with First Class Honours.
- Involved with the Mathematical Society.
- **Selected Modules:** Advanced Quantum Theory, Fluid Mechanics, General Relativity, Superstrings.

New College Pontefract

A-Levels

Pontefract, West Yorkshire

Sept 2019 - June 2021

- A* in Mathematics, A* in Further Maths, A in Chemistry, AS Music.
- Awarded with Certificate of Outstanding Effort and Performance in Further Maths.

Kettlethorpe High School

GCSEs

Wakefield, West Yorkshire

Sept 2014 - June 2019

- 11 qualifications (Levels 9-6).
- Includes Mathematics 9, Physics 8, Chemistry 8, Music 9, A* with Distinction in Further Maths.

University Projects

Chalkdust magazine & Chalkboard Ultra podcast

Mathematics outreach and communication

Oct 2023 - Current

- Writing and editing for a popular maths magazine, distributed nationally among universities.
- Hosting a podcast bringing light to all areas of mathematics, with conversations among PhD students and lecturers.

The Standard Model from a Symmetry Perspective

MMath research project

Oct 2024 - May 2025

- Researched the construction of the Standard Model of Particle Physics; studied gauge theory and spontaneous symmetry breaking.
- Deepened my understanding through reading academic papers and lecture notes from various sources.
- Produced a report of my research, alongside an academic poster and presentation, available to view on my website.

On Modelling Seismic Waves

Mathematical Modelling module

Dec 2022 - Jan 2023

- Conducted research into how we can simulate the effects of seismic waves to better understand the Earth's structure.
- A solo project, programming 2-dimensional partial differential equation solvers into Python.
- Produced a typeset essay to display my findings, receiving a 1:1 (77) in my Mathematical Modelling module.

Skills

Programming $\text{\LaTeX}$ (Overleaf/TeXShop/TeXStudio), Python, HTML, Markdown, Maple, Desmos, VisualPDE.

Soft Skills Teamwork, problem solving, organisation, data analytics, critical thinking, report writing, presentation skills.

Miscellaneous Microsoft Excel, Canva, sleight of hand and card magic.

Volunteering

Stephenson College JCR

Maths subject representative

- Point of guidance contact for students regarding mathematical sciences.
- An advisory role for those needing help with exam techniques, module choices, and general academic problems.

Stephenson College

Jan 2025 - Current

Butler Jazz & Butler Soul Band

President

- Leading two bands after being the lead pianist for two years and Librarian for one year prior.
- Directing rehearsals, booking gigs, organising charity events, overseeing the executive team, managing band finances.
- Improved skills in performance, public speaking, organisation and networking with other bands.

Josephine Butler College

Sept 2023 - Nov 2024

Work Experience

Durham University

Undergraduate tutorial leader

- Assisting undergraduate students with their studies in the form of tutorials; guiding them through chosen problem sheet questions and being a mentor.
- Teaching assistant for Fluid Mechanics III; tutorial leader for Mathematical Methods II and Methods of Mathematical Physics II.

Durham, UK

Oct 2025 - Current

Island Lake Camp

Summer camp counsellor

- Mentor and counsellor for children aged 6-16 at a sleep-away summer camp in America.
- Working as a part of the STEM, Outdoor Adventure and Rockshop departments, ensuring children were safe and engaged in their activities.

Starrucca, Pennsylvania

Summers 2023, 2024 & 2025

Stephenson Central

Bartender

- Responsible for managing the bar through promotions, selling, transactions, stock control, and ensuring a smooth operation.
- Cleaned and restocked whilst customer flow was slow.

Durham, UK

Sept 2023 - June 2025

MyTutor.co.uk

Maths and science tutor

- Tutoring in a classroom environment via video call.
- Private tutoring with 100% return business.

Remote

Jan 2022 - June 2025

Institute of Sales Professionals

Data research assistant and analyst

- Supporting the creation, deployment, and analysis of survey and research projects.
- Project of interest: a study with Cranfield University, delivering insights on the impacts of coaching within sales.

Remote

Jan 2022 - Dec 2022

Achievements

'23-'25 **3x College Scholar Award** – Stephenson College

Durham University

'22-'24 **3x Competitor** – 24 Hours of Le Maths

DU MathSoc

2020 **Regional Finals Competitor** – Senior Team Maths Challenge

UKMT

2019 **The Maureen Winderbank Trophy** – KHS Presentation Event

Kettlethorpe High

2019 **Young Award for Music** – KHS Presentation Event

Kettlethorpe High

2018/19 **Head Boy** – Leadership role

Kettlethorpe High

Interests

Outreach Associate member of the Institute of Mathematics and its Applications (IMA).

Fitness Regularly attend the gym and enjoy running; a trained red belt in Tae Kwon-Do.

Performing Regular at Durham Jazz Society's jazz jam nights @ Fabio's bar.

Scouting Graduated through to Explorer level, began training to become a Young Leader.

Music Worked towards ABRSM Grade 7 classical piano, played the guitar since age eight.

References available upon request.

Last updated 30 November 2025.